Group Number:
Student Names:
Marks: 4 points

COMP 251: Data Structures & Algorithms
Group Assignment 8: Maps and Hash Tables

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*********** INSTRUCTIONS ***************

- Ensure that the outputs of programs match with the format specified in **Question 1**.
- Make sure to add clear screenshots of outputs of your programs in **Question 1** and **Question 2**.
- **Question 3 is optional.** Attempt it only if you wish to further practice Hash Table concepts.
- All submitted work must be your own (or your group's) original work. Academic integrity policies apply.

Question 1: Create an Unsorted Map

Gather all the required interfaces and classes (including the abstract classes) for implementation of an `Unsorted Map`. Note that we studied about the required interfaces and classes during lecture. After gathering the required interfaces and classes, complete the `main` method (given below) so that this method is able to provide an output shown below.

```java
public class UnsortedTableMapExample {
    public static void main(String[] args) {
        // Create a map using the interface type
        Map<String, Integer> map = new UnsortedTableMap<>();

        // Add entries to the map
        map.put("Alice", 25);
        map.put("Bob", 30);
        //.
        //.
        //.
        //****Your code should come here*****

        // Use Iterator, for-each loop, and print keys and values
    } // end main method
} // end class
```

The output should look like this:

Output:

Age of Alice: 25
Age of Bob: 30
Age of Charlie: 35
Removed Bob's age: 30
Is the map empty? False
Entries in the map:
Key: Alice, Value: 25
Key: Charlie, Value: 35
Key: Eva, Value: 40
Size of the map: 3

Question 2: Hash Table with Separate Chaining

Gather all the required interfaces and classes (including the abstract classes) for implementation of a `Hash Table` with separate chaining as a collision-resolution strategy. Note that we studied about the required interfaces and classes during lecture. After gathering the required interfaces and classes, modify the `main` method of question 1 so that this method is able to provide a same output (the order of entries may change) as in question 1.

Question 3: [Optional] Hash Table with Separate Chaining and Linear Probing

(i) Draw the 11-entry hash table that results from using the hash function, $h(k) = (3k + 5) \bmod 11$, to hash the keys 12, 44, 13, 88, 23, 94, 11, 39, 20, 16, and 5, assuming collisions are handled by **separate chaining**.

(ii) Solve the problem (i) assuming collisions are handled by **linear probing**.